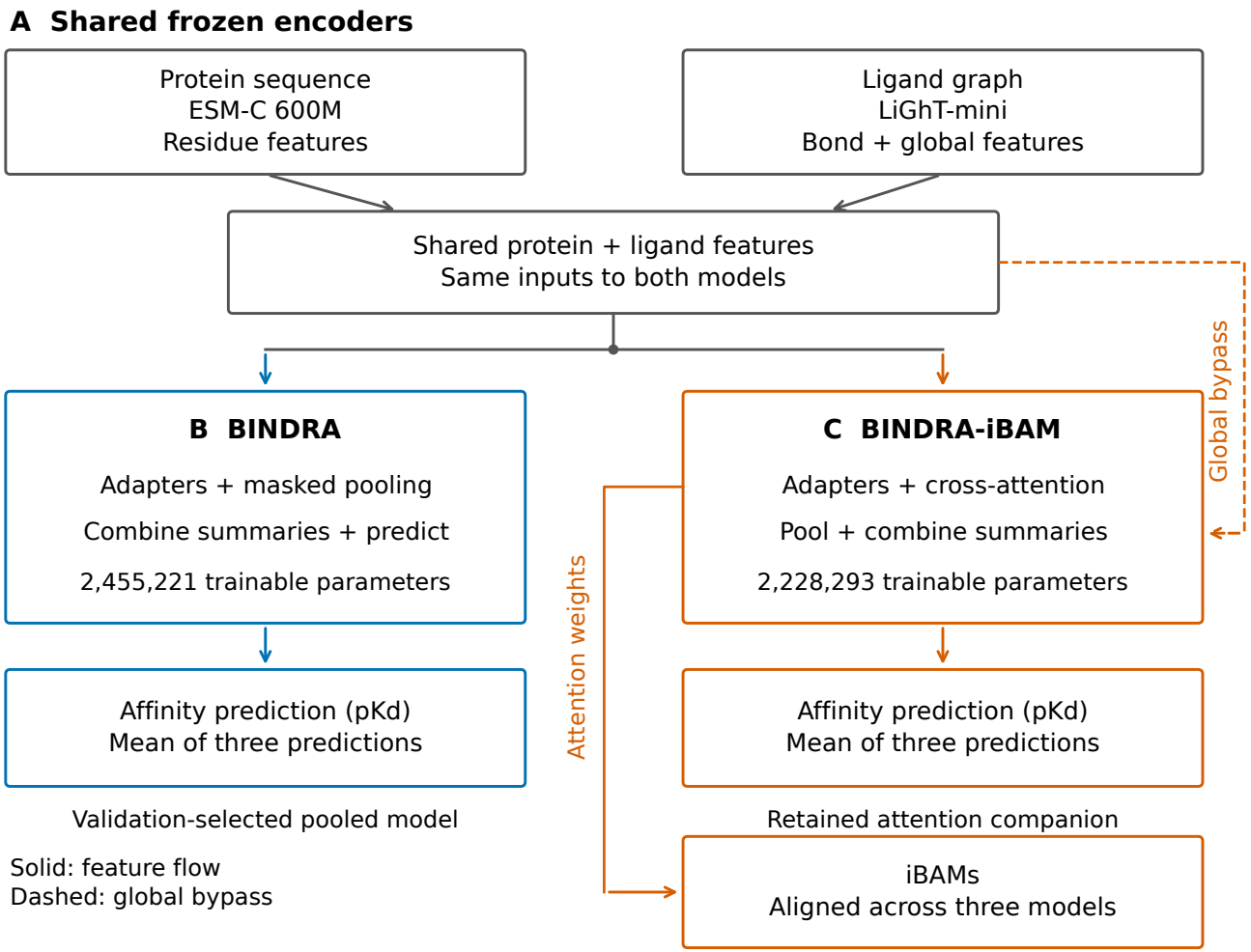
A Shared frozen encoders
Protein sequence
ESM-C 600M
Residue features
Ligand graph
LiGhT-mini
Bond + global features
Shared protein + ligand features
Same inputs to both models
Global bypass
B BINDRA
Adapters + masked pooling
Combine summaries + predict
2,455,221 trainable parameters
C BINDRA-iBAM
Adapters + cross-attention
Pool + combine summaries
2,228,293 trainable parameters
Attention weights
Affinity prediction (pKd)
Mean of three predictions
Affinity prediction (pKd)
Mean of three predictions
Validation-selected pooled model
Retained attention companion
iBAMs
Aligned across three models
Solid: feature flow
Dashed: global bypass